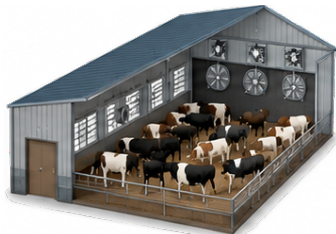
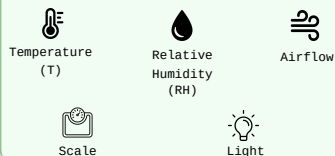


PHYSICAL BARN SYSTEM



Closed cattle-fattening barn
120-head herb

SENSORS (real-time)



ACTUATORS



OUTDOOR WEATHER



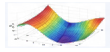
Real-time data

Real-time control

TRANS-DOMAIN DIGITAL TWIN

5-MIN INNER CLIMATE-CONTROL LOOP

Climate Simulator



Evaluate candidates

Candidate Actuator Commands



Daily aggregated data

MPC Decision-Maker

- ✓ Comfort constraints
- ✓ Safety constraints
- ✓ Energy cost
- ✓ Equipment limits
- ✓ Biological guidance

Biological Guidance

Gaussian/Bayesian tuner

DAILY OUTER GROWTH-GUIDANCE LOOP

Daily Aggregation (24h summary)



Growth Simulator (Beef-LiGAPS Layer)



Biological Outputs (daily)



SINGLE-STAGE KNOWLEDGE AND LEARNING LAYER (MEMORY)

CCLL-SEL (Climate-Context Local Library)



Climate contexts & centroids from historical patterns (seasonal, humid, eam, cold, transitional)

SARG-SEL (Stage-Aware Reference-Guided SEL)



Reference growth paths & stage information-> biological guidance for MPC

SS-KStore (Step-Stream Knowledge Store)



Streaming memory of single-episode training (states, actions, rewards, guidance, etc.)

SETD-KStore (Single-Episode Trans-Domain KStore)



- Loadable knowledge package:
- policy snapshots
- Climate contexts
- Bio-guidance-decision links
- Rewards & coverage
- Uncertainty

BIO-AWARE EDGE CONTROLE



Edge Controller

Runs online with knowledge upload, learned offline

Bio-aware MPC command (climate + growth + energy)

Comfort maintained (thermal safety)

Energy-aware operation (lower cost & consumption)

Growth feedback included (feed, heat)

Reusable trans-domain knowledge on edge

Fast control (5-min)

Daily information flow

Biological guidance feedback

Knowledge flow (training & reuse)

SE-TDDT integrates climate, growth, feed response, and energy in a single executable control cycle.

